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Software Requirements Specification

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Field	This example
Team name	5 Idiots
Project title	Pantry Pal Application— A grocery expiration and food waste management application
SDLC chosen in M1 + reason	Incremental — develop the core grocery tracking and expiration alert features first, then gradually add features such as shopping lists, and food waste tracking.
Date / version	v1.0 · Week 3 worked example
GitHub repo	https://github.com/Dora7258/Pantry-Pal.git

1. Introduction

1.1 Purpose and scope

The purpose of this application is to help users manage their groceries and reduce food waste by keeping track of food items and their expiration dates. Users can add grocery items, record their expiration dates, and receive notifications before items expire. The application also helps users identify which groceries should be used first, so they can avoid forgetting food in the refrigerator or pantry, reduce unnecessary spending, and minimize food waste. The initial version focuses on grocery tracking, expiration alerts, and inventory management, with additional features such as recipe suggestions and food waste tracking added later.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Household member	primary user	View and manage shared grocery items and know which items should be used first	Accidentally deleting or changing another person's grocery information
Application administrator	institution supplier or	Manage the application, monitor system performance, and maintain user data	Data loss, system errors, or unauthorized access to user information

2. Overall description

2.1 Product context — what it does / does not

The Grocery Management Application helps users keep track of their groceries and expiration dates. Users can add, edit, and delete grocery items, record expiration dates, receive alerts before items expire, and view items that need to be used soon. The MVP focuses on preventing users from forgetting about groceries and reducing food waste. Additional advanced features are outside the scope of the MVP.

IN SCOPE (MVP this term)		OUT OF SCOPE (explicit promise)
FR-1 — Create / edit / delete account FR-2 — Add / edit / delete grocery items FR-3 Expiration date reminders / notifications FR-4 — Display active Items FR-5 — Show items that expire soon and already expired		Group chat or messaging Online grocery shopping or payment Automatic grocery delivery Integration with supermarkets Advanced AI food recognition Automatic detection of expiration dates from photos Nutrition or calorie tracking Medical/dietary recommendations

2.2 Assumptions and constraints

- **Assumption:** Users have a smartphone and can access the application.
- **Assumption:** Users will enter accurate grocery information, especially the expiration date.
- **Constraint:** The application supports basic grocery information such as item name, quantity, purchase date, and expiration date.
- **Constraint:** The system will not automatically determine whether food is spoiled; it only tracks the information provided by the user.
- **Constraint:** The project must be completed within the assigned development period and available team resources.

3. Functional requirements

As a <role>, I want <goal>, so that <benefit>.

FR-1 — Create / edit / delete account

User story: As a user, I want to create, edit, and delete my account, so that I can manage my personal account information.

Pain this traces to: Users need to manage their account information and should be able to update outdated information or remove their account when they no longer want to use the application.

1. Given the user enters the required account information, when they create the account, the account is saved and can be used to access the application.
2. Given an existing account, when the user edits their account information, the updated information is saved and displayed.
3. Given an existing account, when the user chooses to delete it, the account and its associated information are removed.

FR-2 — Add / edit / delete grocery items

User story: As a user, I want to add, edit, and delete grocery items, so that I can keep my grocery inventory up to date.

Pain this traces to: Users may forget what groceries they have purchased, where they stored them, and when their groceries expire, which can cause food to be unused or wasted.

1. Given the user enters a grocery name and requires information such as expiration dates, when they save it, the item appears in the grocery inventory.
2. Given an existing grocery item, when the user edits its information, the updated information is displayed.
3. Given an existing grocery item, when the user deletes it or marks it as done, the item is removed from the inventory.

FR-3 — Expiration alerts

User story: As a grocery shopper, I want to receive an alert before a grocery item expires, so that I can use it before it expires.

Pain this traces to: Users often forget about groceries in the refrigerator or pantry until they are already expired.

1. The system displays the remaining time until expiration.
2. Given a grocery item has an upcoming expiration date, when the reminder period is reached, the system sends a notification.
3. The notification identifies the grocery item and its expiration date.
4. Items that have already expired are still in notification until it has been removed.

FR-4 — Display Active Items

User story:As a user, I want to see all my current groceries in one place, so that I can easily know what I have at home.

Pain this traces to:Users may buy items they already have because they cannot remember what is currently in their refrigerator or pantry.

1. The system displays all active grocery items in the user's inventory.
2. Each item displays important information such as name and expiration date.
3. The user can view which items are closest to expiration.

FR-5 — Show items that expire soon and already expired

User story:As a user, I want to see which groceries are expiring soon, so that I can prioritize using them.

Pain this traces to:Users may use newer groceries first while older groceries remain unused and eventually expire.

1. The system identifies groceries that are approaching their expiration dates.
2. Items closest to expiration are displayed prominently.
3. The user can distinguish between expiring-soon, and expired items.

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-5 (security/privacy) is required for M2 — every SRS must have at least one.

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	The system shall save account and grocery-item information within 2 seconds after the user submits valid information.	Measure the saving time over 3 trials .
NFR-2	usability	A first-time user shall be able to add a grocery item, including its expiration date, within 60 seconds without assistance.	Conduct a timed test with a user who has not used the application before.
NFR-3	reliability	The system shall retain saved account and grocery information after the application is closed and reopened.	Add information, close the application, reopen it, and verify that the data remains.
NFR-4	reliability	A user shall only be able to access and modify their own account and grocery inventory .	Log in as one user and attempt to access another user's information.
NFR-5	availability	The system shall be available at least 99% of the time , excluding planned maintenance.	Monitor system availability over a defined testing period

ID	Kind	Statement (testable)	How we will check
			and calculate the percentage of uptime.
NFR-6	compatibility	The application shall function correctly on supported Android and iOS devices and different supported screen sizes.	Test the application on multiple supported devices and screen sizes.
NFR-7	Notification reliability	The expiration notification shall be delivered on the scheduled reminder date and identify the grocery item and its expiration date.	Set test expiration dates and compare the scheduled notification with the actual notification.
NER-8	maintainability	The system shall use a modular structure so that account, grocery, scanning, and notification functions can be maintained independently.	Review the system structure and verify that the major functions are separated into appropriate modules.
NER-9	data accuracy	The system shall correctly calculate and display the remaining time until a grocery item's expiration date .	Enter test items with known expiration dates and compare the displayed remaining time with the expected result.
NER-10	error handling	The system shall display a clear error message when required or invalid account/grocery information is entered.	Enter missing and invalid information and verify that an appropriate error message is displayed.

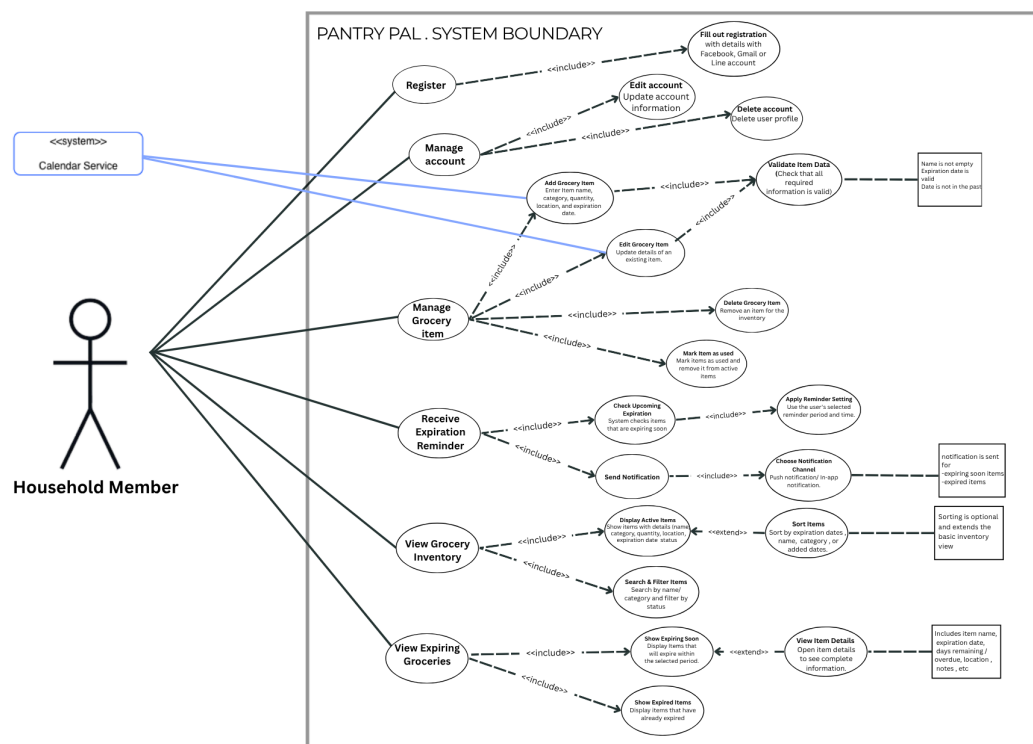
5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Manage account	Household Member	FR-1	The household member creates, edits, or deletes their account, and the system saves or removes the account information accordingly.
UC-2 Manage grocery items	Household Member	FR-2	The household member manually adds, edits, deletes, or marks a grocery item as used, and the inventory is updated accordingly.
UC-3 Receive expiration reminder	Household Member	FR-3	The system detects an upcoming expiration date and sends a reminder to the household member.
UC-4 View grocery inventory	Household Member	FR-4	The household member opens the inventory and sees all currently active

Use case	Actor	Related FR	Brief success story
			groceries, including their names and expiration dates.
UC-5 View expiring groceries	Household Member	FR-5	The household member views groceries that are expiring soon or have already expired and identifies which items need attention.

5.2 Use-case diagram.



6. Out of scope

1. Online grocery shopping or purchasing directly through the app.
2. Payment or subscription features.
3. Automatic grocery delivery services.
4. Nutrition, calorie, or health tracking.
5. Automatic detection of whether food is spoiled or safe to eat.
6. Integration with supermarkets or grocery stores.

7. Sharing or messaging with other users.
8. Scan an Item

7. Traceability matrix (Golden Thread)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Users need to manage their personal information and account securely.	FR-1 Create, edit, and delete accounts.	• User account entity • Registration form • Authentication • Account management (Ch 5- 7)	(Week 6–7)	(Week 7)
Users forget what groceries they have and may waste food or consume expired food.	FR-2 Add, edit, and delete grocery items and mark them as used.	• Grocery item entity • Expiration date field • CRUD operations • Item status (active/used) (Ch 5 & 7)	(Week 6–7)	(Week 7)
Users may forget about groceries until they expire, which can lead to food waste and possible health risks from consuming expired food.	FR-3 Send expiration alerts/notifications.	• Expiration-date checking • Calculate remaining days • Reminder setting • Notification service (Ch 5 - 7)	“Deadline reminder” notification (Ch6)	tasks_due_for_reminder() unit tests (Ch7)
Users may not remember what groceries they currently have at home.	FR-4 Display all active grocery items with details.	• Query active items • Display grocery name and expiration date • Display item status (Ch 5 & 7)	(Week 6–7)	(Week 7)
Users may overlook groceries that are about to expire or have already expired.	FR-5 Show items that are expiring soon and already expired.	• Filter items by expiration date • Identify expiring-soon items • Identify expired items • Sort by expiration date (Ch 5 & 7)	(Week 6–7)	(Week 7)

8. AI usage log

<i>Date</i>	<i>Tool / model</i>	<i>What we asked</i>	<i>What AI produced</i>	<i>What we changed / verified</i>	<i>What we learned</i>
22 Aug 2026 28 Aug 2026 29 Aug 2026	Chat GPT/ Claude/ Perplexity	Help us to modify the requirements and use cases for a Pantry Pal System, including functional requirements, non-functional requirements.	AI suggested FRs, NFRs, and a detailed for managing groceries, expiration dates, reminders, and scanning.	We reviewed the suggestions, changed the actor to Household Member, corrected the FR and use-case relationships, removed unnecessary details, and verified that manual entry belongs to grocery management	We learned how to distinguish FRs from NFRs, connect FRs to use cases, and use include/extend relationships appropriately.

9. Self-Check

- **Every FR is a user story with a benefit** — all 5 FRs include a “so that” clause connected to a real grocery-management problem.
- **Every FR has acceptance criteria you could actually test** — each FR has clear and testable acceptance criteria.
- **No FR is vague** — the FRs use specific actions such as create, add, edit, delete, display, and notify.
- **NFRs exist, including security / privacy** — NFRs cover performance, usability, reliability, security/privacy, compatibility, data accuracy, and error handling.
- **Every FR traces back to a stated problem** — each FR includes a “**Pain this traces to**” statement explaining the problem it addresses.
- **Scope and out-of-scope are stated** — the project scope and features that are not included are clearly defined.
- **Use-case list matches the FR list** — there are 5 use cases, one for each FR, with **Household Member** as the main actor.
- **AI log is honest** — the AI usage log shows what we asked AI to help with and what we reviewed and changed.